



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	CATERCONNECT: AN AGENTIC AI-POWERED CATERING BOOKING AND SURPLUS FOOD REDISTRIBUTION SYSTEM	
S. No.	University ID	Name
1	2420030554	Y. Paavana Pranavasri
2	2420030416	A. Divya Sree
3	2420030392	B. Yaswanth
Name of the Guide		Venkateswari Ketineni

Abstract:

Traditional catering booking systems mainly provide basic functionalities such as searching for caterers, selecting menus, checking packages, booking services, and making payments. However, customers often need to manually compare multiple catering options based on event type, number of guests, budget, cuisine, and availability. The existing catering platforms generally focus on the booking process and do not provide an integrated mechanism for managing suitable surplus food generated after an event. Large events such as weddings, parties, and corporate functions may leave edible food unused, while NGOs and food-serving organizations may simultaneously require food for people in need. The absence of an intelligent connection between caterers with surplus food and organizations requiring food creates an opportunity for avoidable food wastage, which highlights the need for an intelligent catering system that can simplify booking and support effective surplus-food redistribution.

To address this need, the proposed **CaterConnect** system combines Agentic AI with Service-Oriented Architecture to provide intelligent catering booking and surplus-food redistribution. The system uses one central Agentic AI as the intelligent decision-making layer and interacts with independent SOA services through APIs.

The first approach introduces a new technique Agentic AI-Based Conversational Catering Booking, where customers can provide their catering requirements using natural language.

The Agentic AI understands requirements such as event type, number of guests, budget, cuisine, date, and availability and interacts with Catering, Menu, Availability, and Booking Services to recommend suitable catering options. After customer confirmation, the AI invokes the Booking Service to complete the booking. This approach transforms the conventional manual catering selection process into an intelligent conversational booking workflow.

The second approach introduces Intelligent Surplus Food-to-NGO Matching to address the problem of suitable food remaining after catering events. Caterers can register surplus food by providing details such as food type, quantity, location, preparation time, collection deadline, and storage or handling information. The Agentic AI analyzes the available surplus food and compares it with registered NGO requirements, including NGO location, food requirements, required quantity, receiving capacity, availability, and pickup constraints. Based on these parameters, the system identifies the most appropriate NGO for the available surplus food instead of simply displaying a list of nearby organizations. This approach creates an intelligent connection between catering providers with surplus food and organizations capable of utilizing it.

The third approach introduces “Automated Donation Notification and Pickup Coordination” to complete the food-rescue workflow. Once the Agentic AI identifies a suitable NGO, it triggers the Donation and Notification Services and sends a structured donation request containing the available food, quantity, location, collection deadline, and donation details. The NGO can accept or reject the request, and when the donation is accepted, the Pickup Service coordinates the collection and delivery of the food. The complete workflow follows the sequence of surplus food registration, AI matching, NGO selection, notification, NGO acceptance, pickup, and donation completion. The

Agentic AI follows an Observe, Analyze, Decide, and Act workflow, while the SOA services execute the corresponding business operations.

Collectively, these approaches contribute to transforming conventional catering management into an intelligent, efficient, and socially responsible digital platform. These methodologies enhance personalization, automation, food redistribution, and service coordination by leveraging Agentic AI and Service-Oriented Architecture. The outcomes of this proposed system hold promises for simplifying catering booking, reducing avoidable food wastage, facilitating surplus food redistribution to registered NGOs, and creating a measurable social impact for communities in need.

Project Architecture/Project Overview:

